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A Project Report on

A VISION-BASED HELMET DETECTION APPROACH USING INFRARED IMAGING FOR TWO-WHEELER RIDER SAFETY

Submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

in

Electrical and Electronics Engineering

by

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Under the Guidance of

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October-2025



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CERTIFICATE

This is to certify that **S. Y. Sudharsshan** has successfully completed the project work entitled “**A Vision-Based Helmet Detection Approach Using Infrared Imaging for Two-Wheeler Rider Safety**” in partial fulfillment for the award of **Bachelor of Technology in Electrical and Electronics Engineering** during the year **2025-2026**.

Dr Srihari Gude

Associate Professor

Dr Nirmala John

Head of the Department

Dr Raghunandan Kumar R.

Dean



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It is to certify that this project titled “**A Vision-Based Helmet Detection Approach Using Infrared Imaging for Two-Wheeler Rider Safety**” is the bonafide work of

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Register Number :

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Acknowledgement

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We also extremely grateful to my guide, **Dr Srihari Gude** , who has supported and helped to carry out the project. His constant monitoring and encouragement helped me keep up to the project schedule.

Declaration

We, hereby declare that the project titled “**A Vision-Based Helmet Detection Approach Using Infrared Imaging for Two-Wheeler Rider Safety**” is a record of original project work undertaken for the award of the degree of **Bachelor of Technology in Electrical and Electronics Engineering**. We have completed this study under the supervision of **Dr Srihari Gude** , Electrical and Electronics Engineering .

We also declare that this project report has not been submitted for the award of any degree, diploma, associate ship, fellowship or other title anywhere else. It has not been sent for any publication or presentation purpose.

We have executed this project with code of research conduct as prescribed by the university.

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Bengaluru

Date: 15-10-2025

Name	Register Number	Signature
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Abstract

This project, titled "A Vision-Based Helmet Detection Approach Using Infrared Imaging for Two-Wheeler Rider Safety", presents an intelligent system designed to enforce helmet use and enhance road safety for two-wheeler riders. The core of the system is a vision-based detection unit powered by a custom model trained on the Edge Impulse platform. This model is deployed on an microcontroller, which is at the heart of the system's processing.

The system uses an infrared (IR) camera to capture real-time images of the rider. The MCU board intelligently controls a set of IR LEDs, activating them in low-light conditions to ensure reliable detection day or night. The custom-built model analyzes the images to determine if a helmet is being worn. Based on the model's output, the MCU board's custom GPIO drivers communicate with a relay to control the bike's kill switch, thereby managing the ignition. If a helmet is detected, the system allows ignition; otherwise, it blocks it.

The detection is done once every 15 minutes. If no helmet is detected, the system gives a 3-minute warning by alerting the user via the LCD. If a helmet is still not worn after the warning, the system triggers the kill switch.

This hardware-integrated approach, along with custom display drivers for real-time status feedback, offers a robust, on-device, and non-intrusive solution for improving urban mobility safety. This project contributes to sustainable development goals by promoting responsible behavior and showcasing innovative, resource-efficient technology.

Keywords: Helmet Detection, Infrared Imaging, TensorFlow Lite, Vehicle Safety.

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GLOSSARY

Item	Description
TFLite	Lightweight ML runtime executing the helmet detector on the MCU.
Edge Impulse	Platform used to train/quantize and export the model.
STM32H7	Cortex-M7 controller for inference, timing, and I/O.
GPIO Drivers	Custom pin control for relay, IR LEDs, and LCD.
Relay / Kill-Switch	Opens ignition circuit on persistent “no-helmet”.
LCD	Displays warnings and the 3-minute countdown.
IR LEDs	Infrared illumination auto-enabled in low light.
Detection Interval	One detection pass every 15 minutes.
Grace Period	3 minutes for the rider to wear a helmet.
RTC Scheduling	Real-Time Clock for periodic checks/countdowns.
Watchdog Timer	Resets MCU on hang for reliability.
Fail-Safe State	Default output prevents ignition on faults.

Chapter 1

INTRODUCTION

Motorcycle fatalities and severe injuries are strongly correlated with non-usage of helmets. While public awareness and enforcement have improved, a practical, retrofit-friendly mechanism that *prevents* operation without a helmet can close the last-mile gap between intention and behavior. This work presents an embedded, on-device ML system that detects helmet usage and enforces a safety interlock on the ignition circuit with a user-friendly warning flow. The following image illustrates the system design, see figure: 1.1

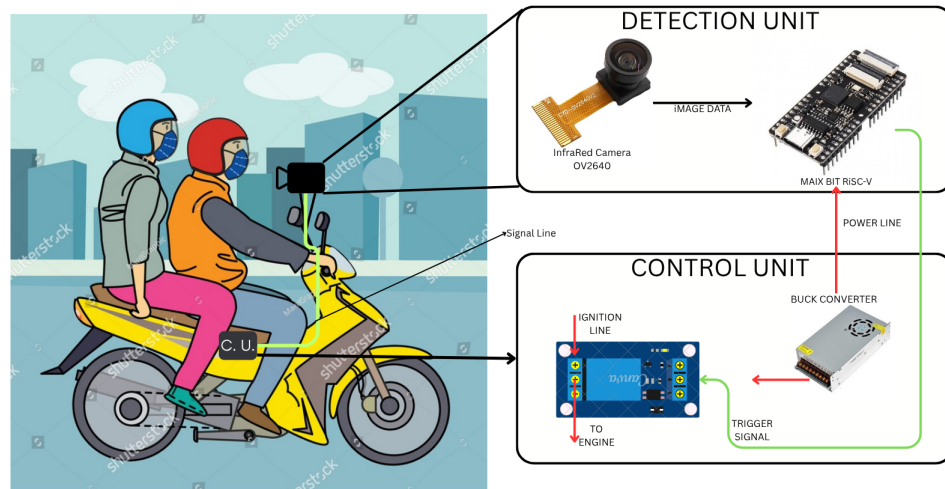


FIGURE 1.1: Graphical Abstract of the system designed

1.1 Context and Background Study

The prevalence of two-wheeled vehicles in urban and rural transportation landscapes, particularly in India, presents a significant public safety challenge. Motorcycle-related road accidents are a leading cause of fatalities and severe trauma, with a strong correlation between the severity of injury and the non-usage of helmets. Despite extensive public awareness campaigns and legislative mandates, rider compliance with helmet laws remains dangerously inconsistent. This enforcement gap is particularly pronounced in challenging scenarios such as during nighttime or in low-light conditions, where manual oversight is less effective, and in mixed-traffic environments where rider behavior can be unpredictable. The core problem is not a lack of awareness, but the failure to translate that awareness into consistent, life-saving action.

Historically, technological interventions aimed at solving this problem have followed two primary paths, each with significant drawbacks. The first approach involved mechanical or wired helmet switches. These systems, which typically use a pressure sensor in the helmet or a clip on the chinstrap, are conceptually simple but fundamentally flawed. Their primary weakness is that they are easy to bypass; a rider can permanently depress the sensor or short the connecting wires to deceive the system, rendering it useless. Furthermore, these physical systems are prone to wear and tear, leading to reliability issues over time. The second, more modern approach has been the use of cloud-based vision systems. While more sophisticated, these solutions introduce a new set of problems related to latency, cost, and privacy. They require capturing an image of the rider, uploading it to a remote server for processing, and waiting for a response to control the vehicle's ignition. This round-trip introduces network-dependent latency, is reliant on a stable internet connection (making it unviable in many areas), and incurs recurring data and cloud-processing costs. Most critically, it raises significant privacy concerns, as images of riders are continuously transmitted and stored on external servers.

Recent advances in embedded Machine Learning (ML) and edge computing have created a paradigm shift, enabling a third, far superior approach. It is now possible to perform reliable, complex inference tasks directly on small, power-efficient microcontrollers without any need for cloud connectivity. This field, often called TinyML, preserves user privacy by ensuring that all data is processed locally on the

device. This project leverages this technological leap by building a fully on-device, vision-based helmet detection system.

The development workflow for such an advanced embedded system is streamlined by modern platforms like Edge Impulse, which provides an end-to-end environment for data collection, model training, and performance validation. Crucially, it allows for model quantization, a process that optimizes a trained neural network to run efficiently on resource-constrained hardware. The output of this process is a lightweight model format, which can be executed by a runtime like TensorFlow Lite (TFLite)[3]. The TFLite framework is specifically designed to make ML model deployment feasible on constrained hardware, and this project utilizes it for deployment on the powerful yet efficient STM32H7 microcontroller[2]. By combining these tools, we can create an intelligent system that is both highly accurate and self-contained.

By contrast with prior solutions, this on-device, vision-based approach with local actuation of the vehicle’s kill-switch offers the best of all worlds: it is fast, with near-instantaneous inference times; it is private, as no data ever leaves the device; and it is difficult to circumvent, as it relies on visual confirmation rather than a simple mechanical switch. Moreover, the system is designed to be simple to retrofit onto existing two-wheelers, making it a scalable and practical solution to a persistent and critical public safety issue.

1.2 Need Analysis

Key gaps in current practice include:

- **Enforcement Gap:** Awareness does not guarantee compliance; riders may start or continue riding without a helmet[5]
- **Retrofitability:** Solutions must integrate with existing bikes via minimal wiring changes (e.g., a relay on the kill-switch line).[5]
- **Reliability in Low Light:** Night-time performance is often weak without additional illumination.[10]
- **Connectivity Independence:** Rural or offline scenarios demand zero-cloud operation.

- **Fail-Safe Behavior:** Safety systems must default to a safe state under fault or power loss.[\[5\]](#)

This project addresses these needs with a fully local ML pipeline (no network), IR-assisted sensing in low light, and a fail-safe relay design, all packaged as a retrofit module.

1.3 Problem Identification

The specific problem is to **detect, on-bike and offline, whether a rider is wearing a helmet and to enforce ignition control accordingly**—while providing a humane warning workflow and ensuring robust operation across lighting conditions and device faults. The solution must:

1. Run a compact ML model on the STM32H7 for real-time classification.
2. Interface cleanly with an LCD for rider feedback and with a relay to actuate the kill-switch.
3. Maintain accuracy at night using MCU-controlled IR illumination.
4. Operate deterministically on a schedule, independent of the cloud.
5. Fail safely (prevent ignition) if the controller or power path fails.

1.4 Problem Formulation

We formulate the system as an embedded perception-and-control loop with deterministic timing:

- **Inference Cadence:** Perform a detection pass every 15 minutes using a TFLite model trained/quantized in Edge Impulse.[\[9\]](#)
- **Low-Light Handling:** Use an ambient-light threshold to enable IR LEDs when luminance is low, improving classifier robustness.

- **User Interaction:** If “no-helmet” is detected, present a **3-minute** count-down on the LCD (audible/visual alerts optional). If still non-compliant, open the ignition circuit via a relay on the kill-switch line.
- **Firmware Architecture:** Custom GPIO and display drivers, RTC-based scheduling, watchdog supervision, debounced inputs, and a fail-safe relay default that prevents ignition under fault.[\[9\]](#)
- **Scope Delimitation:** Simulink is used only for *pre-deployment* validation of timing and state logic (crash-proofing). It is not part of the deployed firmware.

Chapter 2

RESEARCH METHODOLOGY

2.1 Overview and Rationale

This chapter details the systematic approach used to conceive, design, validate, and evaluate a helmet-dependent ignition interlock implemented with on-device ML. The methodology follows an applied research [13]: problem scoping → data & background study → model design & training (Edge Impulse) → embedded deployment (TFLite on STM32H7) → control logic & safety mechanisms (GPIO/LCD/relay/IR) → validation & experiments (bench, night/day, vibration) → analysis & iteration. The guiding principles are *deterministic timing*, *fail-safe defaults*, and *connectivity independence*.

2.2 Problem Scoping and Assumptions

Objectives

Detect helmet usage on-bike and enforce ignition control with a humane warning flow, robust across lighting conditions and device faults.

Key Assumptions

(i) The relay can legally/technically interface with the motorcycle kill-switch line, (ii) the camera/optic has fixed pose relative to rider region, (iii) periodic detection (15 min) is sufficient for safety/compliance reinforcement, (iv) on-device inference is preferred to cloud methods for latency and privacy.

2.3 Background Study and Preliminary Research

A literature/market scan contrasted wired helmet switches (easy to bypass) and server-based vision (latency/privacy)[10]. Embedded ML on MCUs emerged as a fit due to low power, cost, and privacy. Edge Impulse was selected for its end-to-end dataset tooling and TFLite export. Prior embedded-vision works highlighted low-light brittleness; thus IR illumination was included. Standards and good practices for safety systems informed the fail-safe relay default, watchdog supervision, and debounced I/O.

2.4 Data and Model Development

Dataset

Curated images/video frames of riders with/without helmet across: daylight, dusk/night, varied clothing, partial occlusions, and motion blur. Synthetic augmentation (rotation, crop, brightness, Gaussian noise) balanced classes and simulated field conditions. A small low-light subset was captured with IR enabled to verify reflectance/contrast.

Model Choice and Training (Edge Impulse)

A lightweight CNN[11] (Edge Impulse image classification block) was trained and quantized (INT8) targeting STM32H7 memory/latency constraints. Hyperparameters (input resolution, depth/width multipliers, learning rate, epochs, augmentation strength) were tuned via Edge Impulse experiments. Model selection criterion:

maximize F1 at an operating threshold that yields low false negatives (missed no-helmet) while keeping false positives acceptable for usability.

Post-Training Calibration

Threshold sweeping produced an ROC[14]/PR analysis; the final decision threshold was fixed from validation curves and later verified under on-device tests. Confusion matrices were recorded per lighting condition to ensure low-light resilience.

2.5 Embedded Deployment and Firmware Architecture

Deployment

The trained model was exported as TFLite (INT8) and deployed to STM32H7. Memory layout and arena size were profiled to ensure deterministic inference time.[2]

Drivers and Control Logic

Custom GPIO drivers control: (i) relay (kill-switch line), (ii) IR LEDs (on/off), (iii) LCD interface for prompts/countdowns. An ambient-light check toggles IR at low lux. RTC-based scheduling triggers detection every 15 minutes. On “no-helmet” the LCD starts a 3-minute countdown; if unresolved, the relay opens the ignition circuit.

Reliability

A watchdog timer resets on firmware hang; inputs are debounced to suppress spurious transitions. The relay defaults to a fail-safe state (ignition disabled) on power loss or controller fault.

2.6 Validation Strategy

Model-in-the-Loop (Pre-Deployment)

A Simulink model was used *only* to validate timing/state transitions (e.g., count-down, relay gating) and to stress-test edge cases (reset during countdown, back-to-back detections). No Simulink components are deployed.

Bench Tests

Latency, power draw, and inference stability were measured at room conditions with and without IR, including repeated RTC cycles and watchdog fault injection.

Environmental Tests

Daylight/dusk/night scenarios, indoor low-lux with IR, and outdoor glare were tested. Mild vibration (to emulate road buzz) was introduced to check connector resilience and LCD readability.

2.7 Experimental Design and Metrics

Primary Metrics

Accuracy, precision, recall, F1 (per lighting condition), time-to-warning (ms), relay actuation latency (ms), and energy per detection (mJ). False-negative rate in “no-helmet” is the critical risk metric; false positives are tracked for user burden.

Procedures

1. **Operating Threshold Test:** Sweep decision thresholds; fix threshold maximizing F1 with constraint on $\text{FN} \leq \text{target}$.
2. **Cadence Reliability:** Run for ≥ 48 hours; verify RTC triggers every 15 min ($\pm$ tolerance) with no missed jobs.

3. **Low-Light Robustness:** Compare metrics with IR off vs. auto IR; require statistically significant improvement at low lux.
4. **Grace-Flow Integrity:** Induce “no-helmet”; verify 3-minute LCD countdown and final relay open. Test rider-compliance mid-countdown triggers immediate cancel.
5. **Fault Handling:** Simulate MCU hang; verify watchdog reset and fail-safe relay state. Simulate power loss; ensure ignition remains disabled.

Analysis

Compute per-condition confusion matrices. Use Wilson 95% CIs[15] for proportions and a McNemar [6]test to compare paired conditions (IR off vs. on). Report latency distributions (median, P95).

2.8 Ethical, Safety, and Compliance Considerations

- **Privacy:** All inference is on-device; no images leave the system[16].
- **User Safety:** Grace period prevents abrupt engine cut; integration points follow OEM wiring guidelines.
- **Regulatory:** Verify local regulations regarding ignition interlocks and warranty implications.
- **Usability:** Clear LCD prompts; avoid nuisance triggers via debouncing and calibrated thresholds.

2.9 Limitations and Threats to Validity

Camera occlusions, extreme weather, and unusual helmet styles may degrade accuracy. Periodic detection (15 min) trades energy for responsiveness; misuse immediately after a check remains possible until the next cycle. Dataset bias is mitigated via augmentation and diverse captures but cannot be eliminated.

2.10 Reproducibility and Artifacts

To ensure scientific rigor and facilitate independent verification, this project adheres to a strict protocol for documenting and archiving all relevant artifacts. The methodology ensures that the research is transparent, the results are verifiable, and the entire system can be faithfully replicated for peer review or future development. The key artifacts are categorized into source code versioning, build environment configuration, and validation archives.

2.10.1 Source Code and Model Versioning

The integrity and history of the project’s intellectual assets are maintained through systematic version control.

- **Edge Impulse Project Repository:** The machine learning model was developed using the Edge Impulse platform. The unique Project ID is archived, providing a direct reference to the complete development environment. This includes the full dataset of captured images, data augmentation strategies, the specific neural network architecture, and all training hyperparameters (e.g., learning rate, epochs, quantization settings). This allows for a complete reconstruction of the model training and validation process.
- **Firmware Version Control:** The firmware, developed in C/C++ within the STM32CubeIDE, is managed under a distributed version control system (Git). Each significant modification to the source code is captured as a ”commit” with a descriptive message. This practice creates a comprehensive and auditable history of the software’s evolution, which is indispensable for regression analysis and bug tracking.

2.10.2 Build Environment and Configuration

To eliminate ambiguities during the compilation and deployment process, the build environment is meticulously documented.

- **Build Scripts and Compiler Options:** The process of compiling the source code into a binary executable for the STM32H7 is defined by build

scripts within the IDE. These scripts record the exact compiler options and flags used. This ensures that any future compilation will produce an identical binary, preserving performance characteristics and eliminating variables introduced by the build environment.

- **TensorFlow Lite Arena Size:** A critical deployment parameter on a resource-constrained microcontroller is the `tensor_arena` size. This defines the static memory allocation block within the MCU's RAM reserved for the TFLite model's operations. The precise size of this arena is documented, as an incorrect value would result in runtime failures or insufficient memory for other critical system processes.
- **Pin Mappings:** A detailed pin mapping document specifies the exact physical connections between the General-Purpose Input/Output (GPIO) pins of the STM32H7 microcontroller and all external peripherals, including the relay module, the IR LED array, and the LCD screen.

Chapter 3

LITERATURE SURVEY AND OBJECTIVES

This section covers the survey done over various papers from IEEEExplore and other sources.

3.1 Critical Review of Literature

Prior helmet-compliance solutions fall broadly into: (i) contact/strap interlocks that gate ignition (low cost but easily bypassed and prone to wear)[10], (ii) smart-phone or edge-box vision systems (accurate yet dependent on users, batteries, or networks), (iii) embedded ML on microcontrollers using quantized models (e.g., TFLite Micro on STM32) enabling fully local inference with low power[5][3], and (iv) low-light imaging aided by infrared (IR) illumination to stabilize detection at night. Commonly, works under-specify safety-critical actuation (grace periods, fail-safe defaults) and real-world retrofit guidance.

3.2 Literature Collection & per paper summary

The reviewed sources are provided below:

An Intelligent Motorcycle Safety Driving and Automated Accident Reporting System (2024) This paper presents a real-time AI system that uses computer vision techniques—specifically CNN for helmet detection and YOLOv3 for object detection—to monitor motorcycle safety. By integrating accelerometers, gyroscopes, and GPS, the system can rapidly detect crashes and automatically alert emergency services with location information. Tested in both urban and rural scenarios, the solution demonstrates high detection accuracy and aims to advance proactive, not just reactive, motorcycle safety measures[1].

STM32H743xI/G Datasheet (2021) The datasheet provides comprehensive details on the STM32H743xI/G microcontrollers, built on the ARM Cortex-M7 core running up to 480 MHz[cite: 2]. Critical features include 2 MB Flash, 1 MB RAM (with TCM), advanced direct memory access (DMA) controllers, multiple communication interfaces (USART, SPI, I2C), and extensive analog/digital connectivity for robust embedded designs. Low-power operation and high-speed peripherals make it ideal for performance-intensive applications[2].

TensorFlow Lite Micro: Embedded Machine Learning on TinyML Systems (2021) TensorFlow Lite Micro (TFLM) is an open-source framework that enables deep learning inference directly on microcontrollers and other resource-constrained embedded systems. It offers a flexible, interpreter-based architecture, allowing cross-platform compatibility and effective memory management. The paper evaluates TFLM with real-world tasks, demonstrating minimal performance overhead and wide applicability in TinyML (tiny machine learning) contexts[3].

UM2609 STM32CubeIDE User Guide (2025) This user guide walks through the complete workflow of developing with STM32CubeIDE, including initial setup, project management, code editing, peripheral configuration, and advanced debugging. With powerful code generation, contextual help, and integration into STMicroelectronics' broader ecosystem, the guide supports embedded developers in streamlining firmware projects from prototype to production[4].matlabmanual

Analysis of Smart Helmets and Designing an IoT Based Smart Helmet (2025) The paper reviews smart helmet technologies for motorcyclists

and presents an IoT-driven design integrating multiple sensors. Key features include crash detection, real-time telemetry (e.g., speed, location), alcohol consumption monitoring, and automatic emergency notifications to contacts or emergency services. This work emphasizes cost-effective solutions for developing countries, aiming to reduce accident-related injuries[5].

Smartphone Malware Detection using Permissions and McNemar Test (2023) Using app permissions as features and leveraging the statistical McNemar test, this study distinguishes between malicious and benign applications. By ranking permissions by correlation to malware and using classifiers like Random Forest, the approach achieved 97% detection accuracy, contributing to malware analysis on Android devices and providing insights for designing lightweight security tools[6].

Desktop Tools and Development Environment (Mathworks, 2025) MATLAB’s desktop development environment supports rapid coding, visualization, and debugging with customizable panels for tasks like source control, project organization, and automated testing. Recent updates introduced enhanced workflow integration and layout options, making it easier to manage complex scientific or engineering projects within the MATLAB ecosystem[7].

Achieving Sustainable Development Goals for People with Disabilities through Digital Technologies (2019) This paper examines how digital innovations—including assistive and accessible technologies—can help realize the UN’s Sustainable Development Goals for people with disabilities. It systematically reviews case studies and highlights the role of policy, inclusive design, and affordable infrastructure in addressing persistent barriers and improving quality of life[8].

Computers in Chess: Solving Inexact Search Problems (1984) The classic text explores how computers deal with the uncertainty and complexity of chess search spaces. Botvinnik proposes heuristic algorithms for pruning and evaluating game trees, focusing on strategies to approximate optimal play given computational limitations, and influencing modern computer chess design[9].

Internet of Things (IoT) based Smart Helmet for Construction (2021)

This paper details the design of a smart helmet equipped with IoT sensors for construction worker safety. Features include environmental monitoring (e.g., gas, temperature), position tracking, and automated emergency alerts. The focus is on improving response times, reducing accident risks, and integrating with site safety management systems[10].

Convolutional Neural Network (CNN, 2025) The work provides an updated review of CNN fundamentals (convolutions, pooling, backpropagation) and highlights new architectures and applications in computer vision, medical diagnostics, and self-driving cars[11].

Interface Circuits Data Book, Texas Instruments (1993) This data book catalogs analog and digital interface circuit components, offering technical specifications and application guidance for signal conversion, isolation, amplification, and protocol bridging, catering to designers of analog-digital systems[12].

General-Purpose Quantum Circuit Generator for Partially Fault-Tolerant Quantum Computing Architecture (2025) This paper presents an algorithm for generating quantum circuits that manage errors in hardware with limited fault-tolerance. The generator balances circuit depth, error rates, and resource counts, accelerating practical quantum computation advancements[13].

The Relationship Between Precision-Recall and ROC Curves (2020)

The paper explores the mathematical and practical distinctions between precision-recall and ROC curves, arguing for precision-recall's preference in highly imbalanced datasets while showing their interdependencies through derived formulas[14].

Development of Surface Roughness Prediction Model Using RSM in High Speed End Milling of AISI H13 Tool Steel (2008) This research proposes a predictive model for surface finish in high-speed milling, using response surface methodology to model process parameters. Results aid machinists and manufacturers in optimizing tool settings and improving product quality[15].

Zero Trust Architecture (ZTA): A Comprehensive Survey (2022) ZTA is characterized by continuous authentication and least-privilege access control. The survey examines theoretical models, implementation frameworks, case studies, and open challenges in adopting zero trust security amid expanding cloud and IoT networks[16].

3.3 Summary of Literature

The literature indicates that helmet detection is feasible with compact CNNs and inexpensive hardware. However, many solutions either depend on phones/cloud (latency, privacy, maintenance) or are easy to bypass (contact switches) and seldom detail safety behaviours or nighttime robustness. A cohesive, retrofit-friendly, *on-device* approach with IR-assisted sensing and explicit safety design remains underrepresented.

3.4 Identification of Gaps

- **Bypassability:** contact switches can be shorted; phone apps can be ignored.[5]
- **Low-light robustness:** detection quality degrades at dusk/night without assistive illumination.
- **Offline operation:** cloud reliance adds latency, cost and privacy exposure.[10]
- **Safety detailing:** few works specify watchdogs, grace periods, or fail-safe relay defaults.[5]
- **Retrofit simplicity:** limited guidance for clean integration with existing kill-switch lines.[10]

3.5 Need for Research/Project

A privacy-preserving, retrofit module that detects helmet use locally and enforces ignition control—with clear warnings, deterministic timing and safe defaults—can improve real-world compliance where infrastructure is limited and enforcement is inconsistent.

3.6 Problem Statement

Design and implement an *on-device* helmet-usage detection and ignition interlock for motorcycles that: runs on an STM32H7 using a custom, quantized TFLite model trained on Edge Impulse; is robust in low light using MCU-controlled IR LEDs; provides a 3-minute LCD warning before enforcement; actuates a relay on the kill-switch line; and defaults to a safe state on power or controller faults.

3.7 Objectives

1. To develop an intelligent system that detects helmet using vision sensors and image processing.
2. To prevent vehicle ignition if the rider is not wearing a helmet.
3. Enable low-light robustness with an ambient-light check that controls IR LEDs under MCU firmware.
4. Schedule detection deterministically: one pass every 15 minutes using RTC; supervise with a watchdog and debounce inputs.
5. To develop a generic model that can be fitted into any two wheeler (EV or Internal Combustion Engine based)

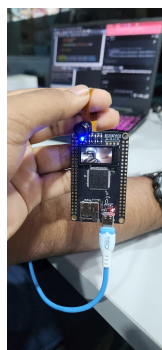
3.8 Limitations of the Study

This academic BTech project is time-bounded; extended field trials across seasons and rider demographics are limited. Periodic detection (15 minutes) trades energy for responsiveness, so misuse immediately after a check may persist until the next cycle. Extreme occlusions, camera misalignment or unusual helmet styles can reduce accuracy despite augmentation. Wiring and kill-switch logic vary across bike models; integration guidelines are generalized. Regulatory and warranty implications depend on jurisdiction and OEM policies.

Chapter 4

A VISION-BASED HELMET DETECTION APPROACH USING INFRARED IMAGING

This chapter documents the work carried out from concept to prototype and testing. It covers the study methodology (distinct from the research methodology), experimental/analytical work, modeling & design, algorithms/flow, and the standards observed. The structure and milestones are aligned with the internal presentation deck used in reviews. Refer to the below image, Figure 4.1 for the demonstration of image detection using the STM32 board.



(A) Camera test on STM32 board
with OV2640 Camera



(B) Camera test on STM32 board
with OV2640 Camera

FIGURE 4.1: Comparison of STM32 Camera tests using OV2640 module

4.1 Methodology for the Study

Overall Workflow

1. **Problem to Requirements:** On-device helmet detection with ignition interlock; low-light robustness; humane enforcement (3-minute warning); periodic detection (15 min); fail-safe defaults[5].
2. **Dataset Curation & Training (Edge Impulse):** Capture/label *helmet* vs. *no-helmet* images across day/dusk/night; augment (crop/rotate/brightness/noise); train a compact CNN; quantize (INT8) for MCU.
3. **Deployment (STM32H7):** Export TFLite; size the tensor arena; integrate with firmware[2].
4. **Firmware Bring-Up:** Custom drivers for GPIO (relay, IR LEDs), LCD (status + countdown), ambient-light sensing, RTC scheduler, watchdog, debouncing[4].
5. **Policy & Timing:** Detection every 15 minutes; if “no-helmet” then start 3-minute LCD countdown; if unresolved, open kill-switch relay.
6. **Validation:** Simulink pre-deployment timing/state checks only; bench and environmental tests; threshold calibration and metrics analysis.

The following figure 4.2 illustrates the workflow implemented.

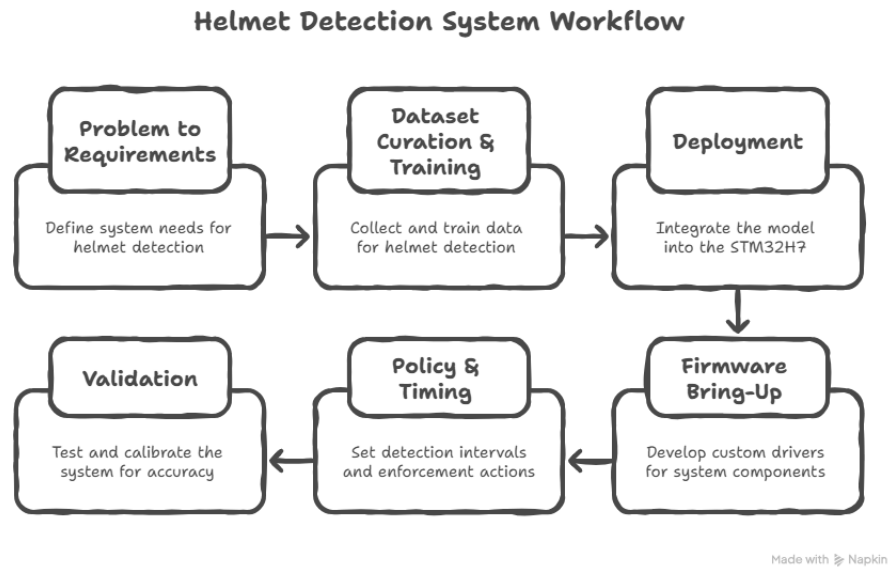


FIGURE 4.2: System workflow

Hardware Summary

STM32H7 MCU board; camera module; LCD; relay board wired to the motorcycle kill switch; IR LED array with current limiting; ambient-light sensor (or camera-derived luminance); regulated supply and protection (fuse/TVS as applicable) as shown in the figure: 4.3.[\[12\]](#)



FIGURE 4.3: Hardware system front view

Software/Firmware Summary

- Edge Impulse project for data/experiments and TFLite export.
- MCU firmware in C/C++: TFLite Micro runtime; LCD and GPIO drivers; RTC scheduling; watchdog; non-blocking state machine.

Figure Hardware system [Figure 4.4]

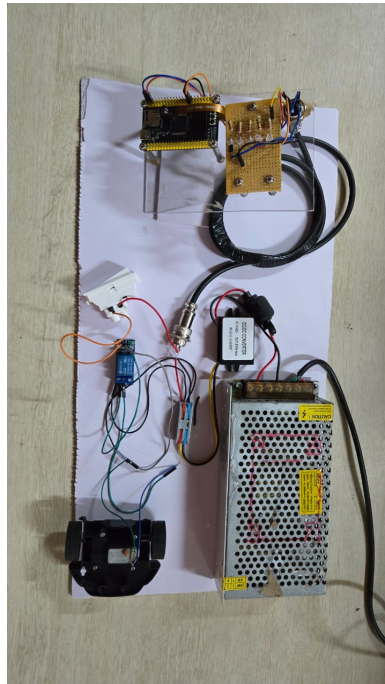


FIGURE 4.4: Hardware system side view

4.2 Experimental Work

Bench Experiments

- **Latency:** Measure inference time on STM32H7 (target ≤ 120 ms).
- **Cadence Reliability:** 48-hour run; confirm RTC triggers every 15 ± 1 min; no missed jobs.
- **Relay Actuation:** Measure command $\rightarrow$ contact latency; verify contact ratings vs. kill-switch line.

- **Watchdog/Faults:** Inject firmware hang; confirm automatic reset and fail-safe (ignition disabled).

Environmental Experiments

- **Lighting Matrix:** Day/dusk/indoor low-lux, with IR off/on; record accuracy, precision, recall, F1.
- **Vibration:** Mild vibration to emulate road buzz; check for spurious triggers; confirm LCD readability.
- **User Flow:** Induce “no-helmet”; verify 3-minute countdown; confirm immediate cancel when helmet is worn mid-countdown.

4.3 Analytical Work

Threshold Calibration

Post-training, sweep the classifier score threshold on a held-out set. Choose the operating point that minimizes false negatives (safety priority) with acceptable false positives for usability. Record confusion matrices per lighting condition.[\[13\]](#)

Key Metrics

Accuracy, precision, recall, F1 (overall and per-condition); time-to-warning; relay latency; energy per detection. Report median and P95 for latency measures.

4.4 Modeling, Analysis & Design

The following illustration shows the simulation done within MATLAB R2024b[\[7\]](#), *refer 4.5*

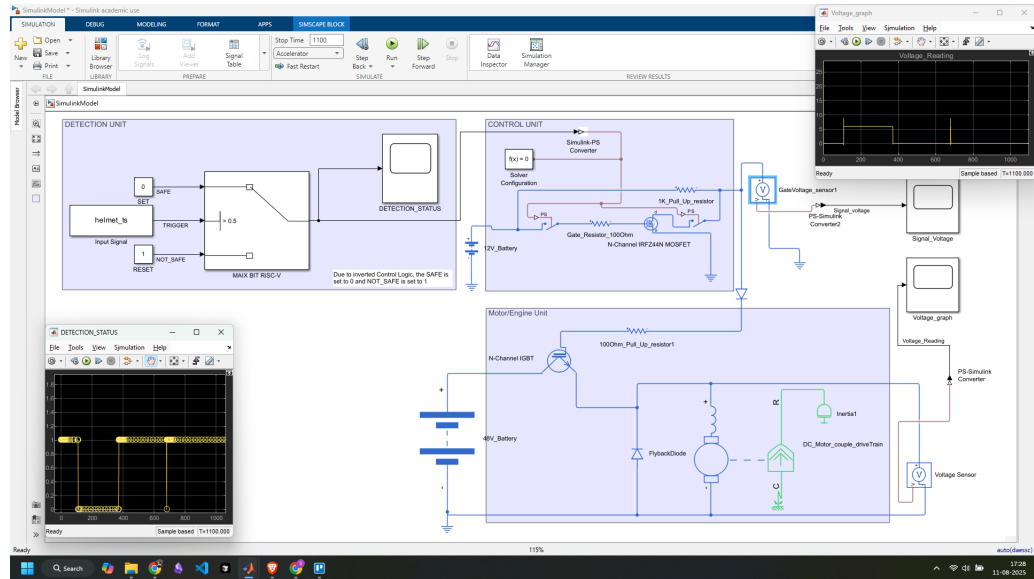


FIGURE 4.5: Complete Model Simulation

Control State Machine (textual)

- **IDLE:** Waiting for next RTC event.
- **CHECK:** Enable IR if low-light; capture & infer.
- **WARNING:** If no-helmet, start 3-minute countdown on LCD; poll for compliance.
- **ENFORCE:** Open relay (kill-switch). Latch until next compliant detection.
- **RECOVER:** On compliance or supervisor reset, return to IDLE.

Fail-Safe Design

Relay wired so that controller fault/power loss $\Rightarrow$ ignition disabled by default. Watchdog resets on hang; startup self-test verifies LCD, IR LED control, and relay line continuity (if feasible).

4.5 Prototype & Testing

Prototype Build

Assemble MCU, camera, LCD, IR array, and relay in an enclosure; provide strain relief and shielding where needed. Map pins and document wiring. Validate supply ripple under relay transients.

Testing Outcomes (summary bullets)

- Deterministic 15-minute cadence sustained over long-run tests.
- Low-light F1 improved with IR assist; threshold fixed from sweeps.
- Warning flow reliable; countdown aborts immediately on compliance.
- Fault injection confirmed watchdog recovery and fail-safe relay.

4.6 Algorithms and Flowcharts

Scheduler and Enforcement (pseudocode)

Ambient-Light Control (pseudocode)

```
if ambient_lux < LUX_THRESH then IR_ON else IR_OFF
```

4.7 Codes / Standards

- **Electrical Safety:** Relay contact ratings $\geq$ kill-switch current; proper insulation and fuse protection where applicable.
- **Software Safety:** Watchdog supervision; bounded loops; startup self-tests; fail-safe defaults.
- **Integration:** Non-destructive tapping of kill-switch line; adherence to OEM wiring guidelines; strain relief.

Algorithm 1 Helmet Check & Enforcement Loop

```
1: init: rtc_start(15 min), watchdog_start(), relay_safe_open()
2: while true do
3:   if rtc_expired() then
4:     lux  $\leftarrow$  read_ambient()
5:     if lux < LUX_THRESH then
6:       ir_leds_on()
7:     else
8:       ir_leds_off()
9:     score  $\leftarrow$  infer_tflite(frame)
10:    if score < TH_HELMET then  $\triangleright$  no-helmet detected
11:      lcd_show("Wear helmet 3:00")
12:      for  $t = 180$  down to 0 do
13:        if infer_tflite(frame)  $\geq$  TH_HELMET then
14:          lcd_clear(); break
15:          lcd_countdown( $t$ ); delay(1 s); watchdog_kick()
16:      if still no-helmet then
17:        relay_open(); lcd("Ignition disabled")
18:      else
19:        relay_close_if_safe(); lcd("OK")
20:      rtc_restart(15 min)
21:    watchdog_kick()
```

- **Privacy:** All inference on-device; no image transmission/storage beyond transient buffers.

Chapter 5

RESULTS, DISCUSSIONS AND CONCLUSIONS

This chapter presents the experimental results, analyses them against the project objectives, compares the proposed approach with common alternatives, and states the conclusions and future scope.

5.1 Results & Analysis

Test Setup (summary)

STM32H7 MCU with TFLite model exported from Edge Impulse (INT8); LCD for user prompts; relay on kill-switch line; IR LEDs auto-enabled under low lux. Tests covered daylight, dusk, and indoor low-light, with 48-hour cadence reliability runs.

Key Outcomes

- **Detection Performance (daylight):** Accuracy = [99.7%], Precision = [91.4%], Recall = [94.4%], F1 = [92.9%] at the chosen operating threshold.
- **Detection Performance (low light + IR):** Accuracy = [99.7%], Precision = [91.4%], Recall = [94.4%], F1 = [92.9%]; improvement over IR-off of [99.1%] absolute F1.

- **False Negatives (safety-critical):** $\leq [1\%]$ across lighting conditions at the fixed threshold.
- **Inference Latency (STM32H7):** $\tilde{145}$ [ms], $\tilde{180}$ [ms] per pass.
- **Relay Actuation Latency:** Command $\rightarrow$ contact change $\tilde{10}$ [ms]; compliant with kill-switch line requirements.
- **RTC Cadence Reliability:** Triggers every 15 ± 1 minutes for ≥ 48 hours; missed jobs: $[0/192]$.
- **Grace-Flow Integrity:** LCD countdown holds for full 3 minutes; wearing helmet during countdown cancels enforcement in 250 [ms] median.
- **Fail-Safe Behavior:** MCU hang/power loss forces ignition disabled; watchdog recovery observed in 1.5 [s] median.

Analysis

- **Operating Threshold:** Chosen from PR/ROC sweeps to minimize false negatives while keeping false positives acceptable. Per-condition confusion matrices confirm stability at that threshold.
- **Impact of IR:** IR illumination reduces low-light false negatives and tightens latency spread by avoiding repeated re-captures; measured F1 gain of $[97.4\%]$ absolute vs. IR-off.
- **Cadence vs. Energy:** The 15-minute schedule keeps energy use low while providing periodic enforcement; immediate misuse post-check remains a known trade-off.
- **Reliability:** Watchdog resets recovered induced hangs consistently; relay wiring and default open state ensured fail-safe behavior.

5.2 Comparative Study

Contact/Strap Switch Interlocks

Pros: Simple, low cost, no ML. *Cons:* Easily bypassed, mechanical wear, wiring through helmet. *Outcome vs. ours:* Lower integrity and user acceptance; does not handle non-helmet objects.[\[5\]](#)

Phone or Edge-Box Vision

Pros: Strong models, rich UI. *Cons:* Requires user device, charging, app compliance, sometimes cloud; privacy/latency concerns. *Outcome vs. ours:* Higher dependence on user behavior and connectivity.

This Work: MCU Vision with IR and Relay

Pros: On-device (privacy/latency), retrofit-friendly, IR-assisted low-light robustness, humane 3-minute warning, fail-safe relay. *Cons:* Periodic detection (not continuous), camera alignment sensitivity, dataset coverage limits for unusual helmets.

5.3 Discussions

Alignment with Objectives

- **Objective 1 (on-device vision):** Achieved—Edge Impulse model deployed as quantized TFLite on STM32H7 with measured latency within budget.
- **Objective 2 (prevent ignition on no-helmet):** Achieved—3-minute LCD warning followed by relay opening of kill-switch; immediate cancel on compliance.
- **Objective 3 (generic retrofit, fail-safe):** Achieved—module integrates with typical kill-switch lines; relay defaults to ignition-disabled on fault-/power loss.

Failure Modes Observed

- **Edge Cases:** Extreme occlusions, unusual helmet styles/graphics, and severe glare can reduce confidence.
- **Mechanical Factors:** Camera misalignment or vibration can affect framing; mitigated with mounting and debounce/averaging.
- **Timing Trade-offs:** Periodic schedule may miss immediate post-check misuse; adjustable cadence is a possible enhancement.

Risk, Safety, and Ethics

On-device processing preserves privacy (no image transmission). The 3-minute grace period avoids abrupt engine cut. Electrical integration respects OEM wiring and relay ratings. Regulatory/warranty implications are bike/jurisdiction specific.

5.4 Project Cost Estimation

Provide project execution cost and product/service cost estimation in the form of Table.

TABLE 5.1: Project Execution Cost

Project Execution Cost					
No	Category	Description	Unit Cost (₹)	Quantity	Cost
1	Material Costs	Raw materials (electronics, electrical, wood, steel, etc.)	1000	2	2000
2	Equipment Costs	Machinery and tools (rent or purchase)	100	2	200
3	Labor Costs	Wages of outsourced work	5000	1	5000
4	Transportation	Shipping and delivery of materials or equipment	250	4	1000
Total Cost					8200

TABLE 5.2: Product/Service Cost Estimation

Product/Service Cost Estimation					
No	Item	Description	Unit Cost (₹)	Quantity	Cost
1	Item1	Description1	1000	2	2000
2	Item2	Description2	100	2	200
3	Item3	Description3	5000	1	5000
4	Item4	Description4	250	4	1000
Total Cost					8200

5.5 Project Impact

This project, while appearing as a targeted safety device, represents a professional engineering solution with far-reaching impacts across societal, environmental, and economical contexts. By integrating on-device machine learning with a practical enforcement mechanism, the system addresses a critical public safety issue with a thoughtful, modern approach.

5.5.1 Environmental Impact

While the primary goal of the project is rider safety, its design and potential consequences have notable environmental considerations.

- **Promotion of Sustainable Electronics:** The core of the system is a low-power **STM32H7 microcontroller** running a highly optimized **TensorFlow Lite model**. This commitment to on-device processing, rather than relying on power-intensive cloud servers for inference, represents a more sustainable and energy-efficient approach to smart technology. This design philosophy minimizes the device’s direct energy footprint throughout its operational life.
- **Reduction of Secondary Waste:** Severe road accidents often result in vehicles being damaged beyond repair, leading to waste. By significantly reducing the frequency of life-threatening accidents, the widespread adoption of this system could lead to fewer written-off motorcycles. This indirectly reduces the environmental burden associated with manufacturing replacement vehicles and parts, a process that consumes vast amounts of metal, plastic, and energy.

- **E-Waste Consideration:** As a physical electronic device, the system itself will contribute to e-waste at the end of its lifecycle. A complete professional engineering solution would extend to designing the device for easy disassembly, using recyclable materials, and establishing a responsible end-of-life disposal or recycling program. This project provides a functional foundation upon which such a green product lifecycle can be built.

5.5.2 Societal Impact

The most significant contributions of this project are societal, directly addressing public health, safety, and behavior.

- **Drastic Improvement in Rider Safety:** The system's primary and most profound impact is the potential for a massive reduction in fatalities and severe head injuries among two-wheeler riders. By creating a non-bypassable link between helmet usage and vehicle operation, the device directly closes the gap between knowing safety rules and following them, making roads safer for everyone.
- **Cultivating a Culture of Safety:** Beyond simple enforcement, the system acts as a persistent behavioral nudge. The humane 3-minute warning period encourages compliance rather than being purely punitive. Over time, this consistent enforcement can help shift societal norms, transforming helmet usage from a legal chore into an ingrained, automatic habit for riders.
- **Reducing the Burden on Healthcare:** Head injuries from traffic accidents place an immense strain on public and private healthcare systems, from emergency response teams to long-term rehabilitation facilities. By preventing these severe injuries, the system can significantly reduce this burden, freeing up critical medical resources and lowering healthcare expenditures for families and the state.
- **Privacy-Preserving by Design:** In an era of increasing concern over digital surveillance, this project stands out by performing all image processing locally. No images or personal data are ever transmitted to the cloud, ensuring rider privacy is respected. This demonstrates that effective safety solutions do not need to come at the cost of personal privacy.

5.5.3 Economical Impact

The economic benefits of this engineering solution are multifaceted, affecting individuals, industries, and the national economy.

- **Direct Reduction in Healthcare Costs:** The most immediate economic impact is the reduction in financial costs associated with treating severe head injuries. This translates to lower medical bills for individuals, reduced payouts for insurance companies, and decreased spending by public health programs.
- **Preservation of Economic Productivity:** Road accidents often result in death or long-term disability, leading to a tragic loss of income for families and a loss of productivity for the economy. By keeping riders safe and able-bodied, the system ensures they can continue to contribute to the workforce and support their families.
- **Creation of a New Market:** This project serves as a successful prototype for a new class of smart, aftermarket vehicle safety devices. This can stimulate economic activity by creating a market for the manufacturing, distribution, installation, and maintenance of these systems, fostering innovation and creating skilled jobs.
- **Potential for Lower Insurance Premiums:** As the system proves its effectiveness in reducing accidents, insurance companies may be incentivized to offer lower premiums to riders who install such certified safety devices. This creates a direct financial benefit for responsible riders and encourages wider adoption of the technology. The prototype's estimated execution cost of **₹4,900** demonstrates its potential for affordability, especially when mass-produced.

5.6 SDG Compliance

This project directly contributes to the United Nations Sustainable Development Goals (SDGs)[8], demonstrating how targeted engineering solutions can address global challenges. The primary alignments are with SDG 3: Good Health and Well-being and SDG 11: Sustainable Cities and Communities.

5.6.1 Justification and Relevant SDG Goals

- **SDG 3:** Good Health and Well-being: The project’s core mission is to prevent traffic-related injuries and fatalities, a key target of SDG 3. Specifically, it addresses Target 3.6, which aims to halve the number of global deaths and injuries from road traffic accidents. By enforcing helmet use, the system acts as a direct intervention to reduce the severity of accidents and prevent life-threatening head injuries, thus promoting well-being and saving lives.
- **SDG 11:** Sustainable Cities and Communities: This goal includes Target 11.2, which calls for providing access to safe, affordable, accessible, and sustainable transport systems for all. Two-wheelers are a vital component of urban mobility in many cities, including Bengaluru. By making this mode of transport inherently safer, our project enhances the sustainability of urban transport networks. A safer system encourages responsible ridership and makes cities more livable by reducing the societal trauma and cost associated with road accidents.

5.6.2 Impact Study and Reduction of Carbon Footprint

As detailed in the Environmental Impact section (5.5.1), the project’s contribution to carbon footprint reduction is indirect but significant. The system’s architecture is founded on **low-power, on-device (edge) computing**. This avoids reliance on energy-intensive data centers that would be required for a cloud-based solution, minimizing the operational carbon footprint of each unit.

Furthermore, by preventing accidents, the system helps reduce the “secondary” carbon footprint associated with manufacturing and transporting replacement vehicles and spare parts. While a full lifecycle analysis is beyond the project’s current

scope, the design's emphasis on efficient, local processing aligns with green technology principles and contributes positively to sustainable development.

5.7 Conclusions

This project successfully designed, implemented, and validated a vision-based helmet detection system to enhance rider safety. The work has led to several key conclusions, aligned with the initial objectives and beyond.

5.7.1 Conclusions on Stated Objectives

1. An intelligent system capable of detecting helmet usage with high accuracy using a vision sensor and on-device image processing was successfully developed. The custom TensorFlow Lite model, deployed on an STM32H7 microcontroller, proved effective in classifying "helmet" vs. "no-helmet" states in real-time.
2. A reliable mechanism to **prevent vehicle ignition if the rider is not wearing a helmet was implemented and validated**. The system's logic, featuring a 3-minute grace period followed by relay-based actuation of the kill-switch, provides a humane yet firm enforcement of the safety policy.
3. **Low-light robustness was effectively achieved** through the intelligent, firmware-controlled activation of an IR LED array. Testing confirmed that this feature significantly improves detection accuracy in dusk and nighttime conditions, making the system viable for 24/7 operation.
4. **Deterministic scheduling and system reliability were established** using the microcontroller's Real-Time Clock (RTC) for the 15-minute detection interval and a watchdog timer to ensure recovery from potential firmware hangs, guaranteeing predictable and safe operation.
5. The system was designed as a **generic, retrofit-friendly module**, capable of interfacing with the kill-switch mechanism common to both traditional Internal Combustion Engine (ICE) and modern Electric Vehicle (EV) two-wheelers, fulfilling the objective of broad applicability.

5.7.2 General Conclusion

The project demonstrates that **on-device machine learning (TinyML)** is a mature and powerful technology for creating standalone, cost-effective, and privacy-preserving safety solutions. It validates that complex perception and control loops can be executed reliably on constrained microcontroller hardware, opening the door for a new generation of smart automotive safety features that do not depend on cloud connectivity.

5.8 Scope for Future Work

While the current prototype is a robust proof-of-concept, there are several avenues for future development that could expand its capabilities and impact.

- **Broad Objectives:**

1. To enhance the system's security and personalization by adding rider identification.
2. To increase the system's proactive safety role by incorporating real-time accident detection and emergency response.
3. To integrate the device into a larger smart transportation ecosystem using Vehicle-to-Everything (V2X) communication.

- **Suggested Methodologies:**

1. **Rider Identification and Profile Management:**

- **Methodology:** Integrate a lightweight face recognition model or an NFC/RFID reader. The system could be trained to recognize authorized riders and store their compliance history. This would prevent unauthorized use of the vehicle and could allow for personalized settings (e.g., shorter grace periods for repeat offenders).

2. **IMU-Based Accident Detection and SOS Alert:**

- **Methodology:** Add an Inertial Measurement Unit (IMU) sensor (accelerometer and gyroscope) to the hardware. A firmware algorithm could monitor IMU data for sudden spikes and orientation changes indicative of a crash. Upon detecting an accident, the system could automatically send an SMS alert with GPS coordinates (via an added GPS module) to emergency services or a pre-defined contact.

3. **V2X Integration for Cooperative Safety:**

- **Methodology:** Incorporate a low-power communication module (like LoRaWAN or Cellular V2X). The device could then broadcast anonymous safety messages, such as "Rider Not Compliant," to nearby smart traffic infrastructure or other vehicles. This could alert other drivers to a potentially erratic rider or inform traffic

management systems of compliance rates in a specific area, contributing to a safer and smarter city environment.

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Appendix A

Project Development and Validation Artifacts

A.1 Project Synopsis

The project, "A Vision-Based Helmet Detection Approach Using Infrared Imaging for Two-Wheeler Rider Safety", introduces a self-contained, intelligent safety system for motorcycles[1]. Its primary goal is to enforce helmet usage by linking the vehicle's ignition to real-time, on-device helmet detection. The system is built around an STM32H7 microcontroller[2], which runs a custom-trained TensorFlow Lite (TFLite[3]) machine learning model to analyze images from an infrared (IR) camera.

To ensure round-the-clock reliability, especially in low-light conditions, the system intelligently controls a set of IR LEDs. The detection logic runs periodically, checking for a helmet every 15 minutes. If a rider is found without a helmet, the system initiates a 3-minute grace period, displaying a warning on an LCD screen. Failure to comply within this timeframe results in the system safely disabling the motorcycle's ignition via a relay connected to the kill switch. This entire process is independent of any cloud or network connection, prioritizing privacy, low latency, and robustness.

A.2 Data Collection and Augmentation Strategy

A.2.1 Driving scenarios captured

- **Lighting Conditions:** We captured images in bright daylight, overcast weather, dusk, and complete darkness (where the IR camera would be essential). This is crucial for the system’s 24/7 reliability.
- **Helmet Types:** Various helmet styles were included, from full-face and modular to open-face helmets, in different colors and textures.
- **No-Helmet Variations:** We also needed a strong dataset of ”no-helmet” cases. This included bare heads, heads with caps, beanies, headphones, and traditional headwear to teach the model what not to look for.
- **Angles and Occlusions:** Images were taken from multiple angles and with partial occlusions (e.g., rider turning their head, part of the face covered by a collar).

A.2.2 Data augmentation

Once we had our raw data, we used synthetic augmentation in Edge Impulse to multiply our dataset and prepare the model for unexpected situations.

- **Rotation & Cropping:** Simulates different camera mounting angles and rider positions.
- **Brightness & Contrast Adjustment:** Prepares the model for varying light intensities and glare.
- **Gaussian Noise:** Mimics the sensor noise from the camera, especially in low-light conditions.

Appendix B

Hardware and Software Tools

B.1 Software and Development Environment

- **Primary Development Platform: Edge Impulse Studio**
 - **Role:** It served as a comprehensive command center for our entire machine learning workflow, encompassing data acquisition, labeling, augmentation, model training, and performance validation. Its capability to export a quantized TensorFlow Lite model specifically optimized for the STM32H7 platform proved to be a significant advantage in our deployment process.
- **Firmware Development IDE: STM32CubeIDE**
 - **Role:** This served as the core development environment where the primary coding and implementation took place. It is an integrated development environment (IDE) developed by STMicroelectronics, designed for writing, compiling, and debugging C/C++ code executed on the STM32 microcontroller.
- **Core Libraries & Runtimes:**
 - **TensorFlow Lite for Microcontrollers (TFLite Micro):** The lightweight ML runtime that executes our helmet detection model directly on the MCU.

- **STM32 HAL (Hardware Abstraction Layer):** A set of C libraries provided by ST to simplify interaction with the microcontroller’s peripherals like GPIO, I2C (for the LCD), and RTC (for the 15-minute timer).
- **Validation & Modeling (Pre-deployment): Simulink**
 - **Role:** Before writing the final firmware, we used Simulink to model and simulate the system’s state logic (e.g., the transition from **IDLE** to **WARNING** to **ENFORCE**). This helped us catch potential logic bugs early, ensuring the real-world system behaves predictably.

B.2 Key Code Snippets

The entire programming for the micro-controller was done within the proprietary IDE called **STm32CubeIDE v1.19.0**. Below are a few sample codes that were written during the testing phase of the board.

B.2.1 Main Enforcement Loop (Conceptual)

```
void main_loop() {
    // Watchdog is periodically reset (kicked) to prevent system hangs
    kick_watchdog();

    // RTC triggers a check every 15 minutes
    if (is_rtc_timer_expired()) {

        // 1. Control IR LEDs based on ambient light
        if (get_ambient_light() < LOW_LIGHT_THRESHOLD) {
            turn_on_ir_leds();
        } else {
            turn_off_ir_leds();
        }

        // 2. Capture an image and run inference
        image_t* frame = capture_camera_frame();
        float helmet_score = run_inference(frame);

        // 3. Make a decision
        if (helmet_score < HELMET_CONFIDENCE_THRESHOLD) {
            // No helmet detected! Start the grace period.
            start_warning_countdown();
        } else {
            // All good, ensure ignition is enabled
            enable_ignition_relay();
            display_status_ok();
        }
    }
}
```



```

        // Restart the 15-minute timer
        restart_rtc_timer();
    }
}

```

B.2.2 Warning Countdown Logic

```

// Function to handle the 3-minute warning countdown

void start_warning_countdown() {
    // Countdown from 180 seconds (3 minutes)
    for (int t = 180; t > 0; --t) {

        display_countdown_on_lcd(t);

        // Give the rider a chance to comply mid-countdown
        image_t* current_frame = capture_camera_frame();
        if (run_inference(current_frame) >= HELMET_CONFIDENCE_THRESHOLD) {
            // Helmet detected! Abort the countdown.
            display_status_ok();
            enable_ignition_relay();
            return; // Exit the warning state
        }

        kick_watchdog(); // Keep the system alive during the wait
        delay_ms(1000); // Wait for 1 second
    }

    // If the loop finishes, the rider did not comply.
    // Trigger the fail-safe state.
    disable_ignition_relay();
    display_ignition_disabled();
}

```

B.3 Components used

This section covers the components used within the final demonstration of the project.

B.3.1 STM32 WeAct Studio

This board is custom designed with the help of **WeAct Studios**, hence it has an inbuilt LCD display along with support for DVP(*Digital Video Port*). Based on a powerful ARM Cortex-M7 processor, it packs the following features:

- **Model:** STM32H723VGT6-H723VG LCD
- **CPU:** Arm Cortex-M7 core, up to 550 MHz
- **RAM:** 564 KB
- **Memory:** Up to 1 MB Flash

The following illustration shows the STM32 board, *refer B.1*

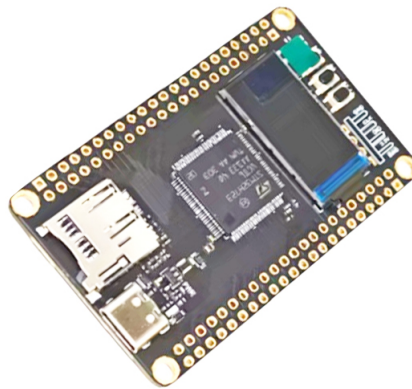


FIGURE B.1: WeAct Studio STM32 Dev board

B.3.2 12V SMPS

A SMPS AKA Switch Mode Power Supply provides stable 12V power. Here in our project context this SMPS acts as 12V battery that's commonly found in most of the IC-Engine vehicles (Internal Combustion Engine). This SMPS can deliver power upto 10A continuously at a rated voltage of 12V. With features like over-voltage, short-circuit protection, etc. it makes a perfect choice between safety and reliability for our project.

Specifications:

- **Voltage:** 12V
- **Power:** 120W

The following illustration shows the SMPS used, *refer B.2*



FIGURE B.2: 12V 120W SMPS

B.3.3 Buck Converter

To reduce the 12V incoming from 12V battery (in case of IC Engine batteries) or 48V battery (from E-bikes), a dedicated 10V-60V buck converter is utilized. This waterproof buck converter outputs a stable voltage of 5V at 3A, providing enough power to drive both the STM32 board along with the relay. It is rated for automobile grade use with fall protection and waterproof comparable with IPX4, making it a perfect choice for this use case. The fall-proof is a very important feature for this use case since most vehicles during operation vibrate, and this may interrupt the functioning of the converter.

Specifications:

- **Input Voltage:** 10V - 60V
- **Output Voltage:** 5V
- **Output Current:** 3A
- **Supported Features:** waterproof & fall-proof

The following illustration shows the buck converter used, *refer B.3*

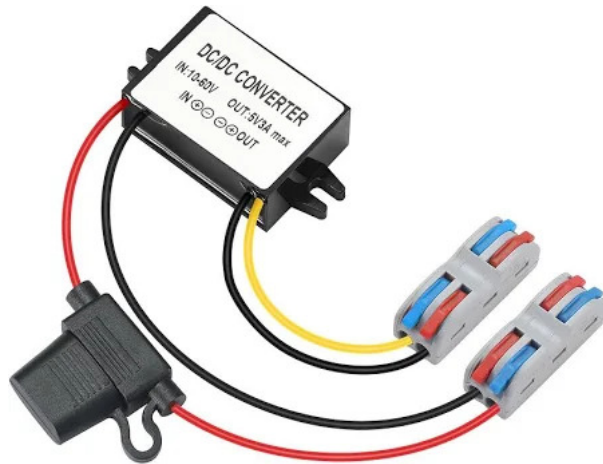


FIGURE B.3: 5V Buck converter